

Campus Event Management System

Technical Architecture & Database Aggregation Report

- **Student Author:** Chamakura Akhil Reddy
- **Date:** September 10, 2026
- **Repository:** <https://github.com/akhill-reddy/campus-event-manager>
- **Tech Stack:** Node.js, Express.js, MongoDB Atlas, EJS, Bootstrap 5

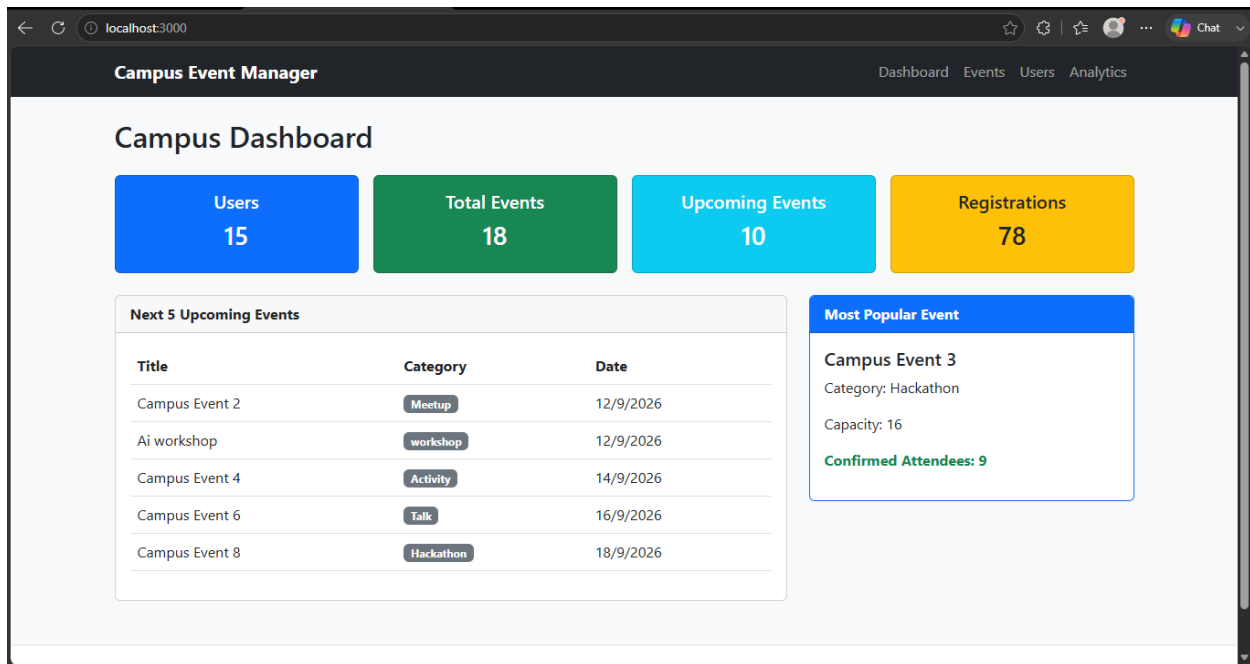
1. Executive Summary & System Architecture

The Campus Event Manager is a multi-tier full-stack application built to streamline university event scheduling, participant tracking, and real-time database aggregations using Node.js, Express, and MongoDB Atlas.

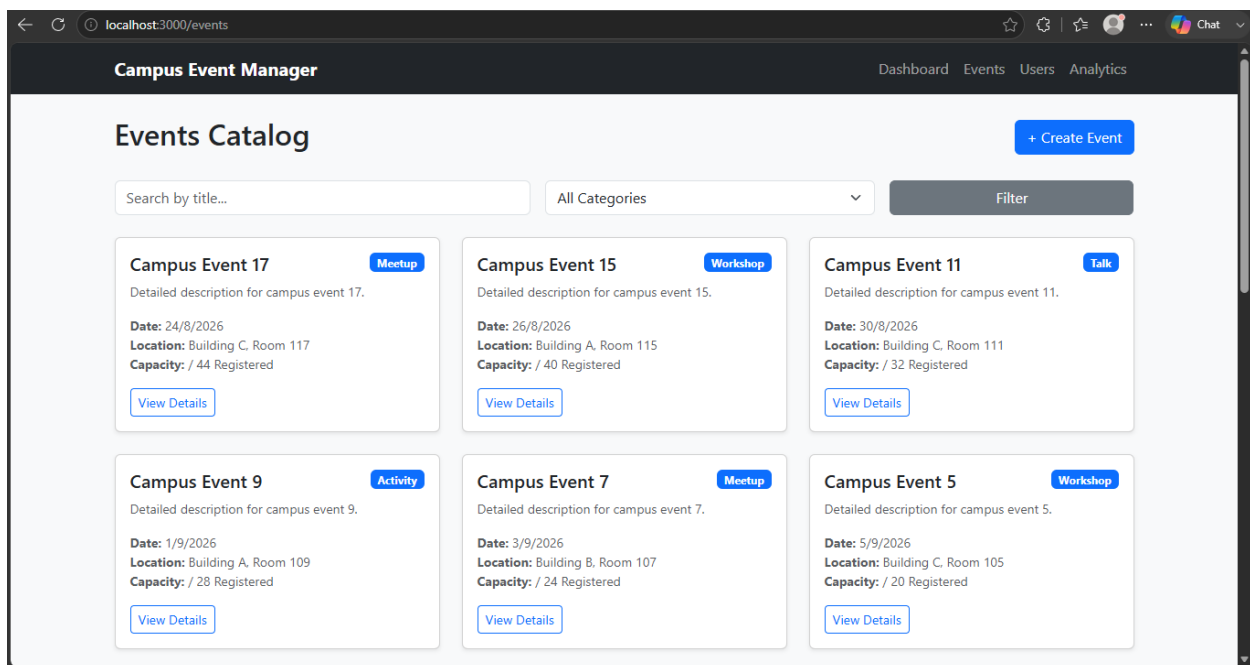
- **Backend Runtime (Node.js / Express.js):** Handles REST routing, middleware request validation, and dynamic view rendering.
- **Database Store (MongoDB Atlas / Mongoose):** Document persistence store managing relational references and embedded participant arrays.
- **Presentation Layer (EJS & Bootstrap 5):** Server-side templating providing responsive layouts and component styling.
- **Analytics Engine (MongoDB Aggregations):** Processes multi-stage pipeline operations (\$unwind,\$group, \$lookup,\$project).

2. System User Interfaces & Modules

- **2.1 Main Operations Dashboard:** Centralized command center featuring live operational counters for total users, active events, upcoming schedules, and popular event tracking.



- **2.2 Events Catalog & Management:** Comprehensive catalog interface with category filtering, search capabilities, and capacity indicators for workshops, meetups, and talks.



- **2.3 Event Details & Participant Registration:** Detailed metadata view providing venue specifics, start/end timestamps, tag mappings, and an interactive participant roster management table.

localhost:3000/events/6a977d04f69f6e6dd8987aa5

Campus Event Manager

DashboardEventsUsersAnalytics

Campus Event 17

Meetup

Detailed description for campus event 17.

Start: 24/8/2026, 12:00:00 am

End: 24/8/2026, 2:00:00 am

Capacity: 44

Location: Building C, Room 117

Tags: MongoDB AI

Register Participant

Select User...

Register User

Registered Attendees (3)

User ID	Registered Date	Action
6a977d04f69f6e6dd8987a86	3/9/2026	Cancel
6a977d04f69f6e6dd8987a87	3/9/2026	Cancel

- 2.4 User Directory & Role Management: Administrative directory mapping user roles (Professor, Researcher, Student) and department affiliations with quick-access history tracking links.

localhost:3000/users

Campus Event Manager

DashboardEventsUsersAnalytics

User Directory

Search by name or email...

All Roles

All Departments

Filter

Full Name	Email	Role	Department	Registrations	Action
User1 Student1	user1@university.edu	Professor	Data Engineering	13 events	View History
User10 Student10	user10@university.edu	Professor	Cybersecurity	0 events	View History
User11 Student11	user11@university.edu	Researcher	Cloud Computing	0 events	View History
User12 Student12	user12@university.edu	Student	Computer Science	0 events	View History
User13 Student13	user13@university.edu	Professor	Data Engineering	0 events	View History
User14 Student14	user14@university.edu	Researcher	Cybersecurity	0 events	View History
User15 Student15	user15@university.edu	Student	Cloud Computing	0 events	View History
User2 Student2	user2@university.edu	Researcher	Cybersecurity	13 events	View History
User3 Student3	user3@university.edu	Student	Cloud Computing	13 events	View History

3. MongoDB Aggregation Pipelines & Analytics

- **3.1 Analytics & Reporting Interface:** Visualizes backend aggregation results directly inside the browser application.

The screenshot shows a web application titled "Campus Event Manager" with a navigation bar containing "Dashboard", "Events", "Users", and "Analytics". The main content area is titled "MongoDB Aggregations & Analytics" and displays three tables:

Analysis A: Registrations by Category		
Category	Events Count	Confirmed Regs
Meetup	4	18
Talk	4	18
Workshop	3	18
Hackathon	3	15
Activity	3	9
workshop	1	0

Analysis C: Users With No Registrations		
Name	Email	Role
User10 Student10	user10@university.edu	Professor
User11 Student11	user11@university.edu	Researcher
User12 Student12	user12@university.edu	Student
User13 Student13	user13@university.edu	Professor
User14 Student14	user14@university.edu	Researcher
User15 Student15	user15@university.edu	Student

Analysis B: Top 5 Most Popular Events				
Event Title	Category	Capacity	Registrations	Occupancy
Campus Event 11	Talk	32	9	28%
Campus Event 7	Meetup	24	9	38%
Campus Event 15	Workshop	40	0	22%

Pipeline 1: Total Registrations by Category

Plaintext

```
db.events.aggregate([
  { $unwind: "$registrations" },
  { $group: { _id: "$category", totalEvents: { $addToSet: "$_id" }, confirmedRegistrations: { $sum: 1 } } },
  { $project: { category: "$_id", eventsCount: { $size: "$totalEvents" }, confirmedRegistrations: 1, _id: 0 } },
  { $sort: { confirmedRegistrations: -1 } }
]);
```

Pipeline 2: Top 5 Events by Occupancy Rate

Plaintext

```
db.events.aggregate([
  { $project: { title: 1, category: 1, capacity: 1, registrationsCount: { $size: { $ifNull:
    ["$registrations", []] } }, occupancyRate: { $multiply: [ { $divide: [ { $size: { $ifNull:
    ["$registrations", []] } }, "$capacity" ] }, 100 ] } } },
  { $sort: { occupancyRate: -1 } },
  { $limit: 5 }
]);
```

Pipeline 3: Inactive Users Query (\$lookup)

Plaintext

```
db.users.aggregate([
  { $lookup: { from: "events", localField: "_id", foreignField: "registrations.userId", as:
    "userEvents" } },
  { $match: { userEvents: { $size: 0 } } },
  { $project: { firstName: 1, lastName: 1, email: 1, role: 1 } }
]);
```

4. Project Verification

- **Developer:** Chamakura Akhil Reddy
- **Repository Link:** <https://github.com/akhill-reddy/campus-event-manager>
- **System Status:** Fully Operational & Verified